

TABLE I: Training data in Prosa

Level	File	Theorem Name
level	PROSA file	Statement name
1	prosa/analysis/abstract/abstract_rta.v	job_is_completed_by_arrival_plus_R
1	prosa/analysis/facts/behavior/supply.v	supply_during_complement
1	prosa/implementation/facts/generic_schedul_e.v	schedule_up_to_prefix_inclusion
1	prosa/model/priority/numeric_fixed_priorit_y.v	NFPD_is_total
1	prosa/results/optimality/edf.v	EDF_WC_optimality
1	prosa/classic/analysis/apa/bertogna_edf_comp.v	bertogna_edf_comp_iteration_monotonic
1	prosa/classic/model/arrival/jitter/arrival_sequence.v	arrived_between_implies_in_actual_arrivals
1	prosa/classic/model/schedule/uni/jitter/schedule.v	service_before_jitter_is_zero
1	prosa/classic/util/div_mod.v	modnSor'
1	prosa/classic/analysis/global/basic/bertogna_edf_comp.v	edf_claimed_bounds_ge_cost
2	prosa/analysis/abstract/IBF/task.v	interference_plus_sched_le_serv_of_task_plus_task_interference_job
2	prosa/results/rta/ideal/fp/bounded_pi.v	A_is_in_concrete_search_space
2	prosa/results/transfer_schedulability/criterion.v	slackless_interval_step_case_completed_job
2	prosa/util/nondecreasing.v	nondecreasing_sequence_undup
2	prosa/analysis/abstract/restricted_supply/bounded_bi/jlfp.v	job_arrival_is_bounded
2	prosa/classic/analysis/apa/bertogna_edf_comp.v	taskset_schedulable_by_edf_rta
2	prosa/classic/model/schedule/apa/constrained_deadlines.v	platform_fp_no_multiple_jobs_of_interfering_tasks
2	prosa/util/div_mod.v	div_ceil_subadditive
2	prosa/classic/implementation/apa/schedule.v	scheduler_apa_work_conserving
2	prosa/classic/analysis/global/parallel/workload_bound_n_k_covers_all_jobs.v	workload_bound_n_k_covers_all_jobs
3	prosa/analysis/abstract/restricted_supply/abstract_rta.v	IBF_NP_bounds_interference
3	prosa/analysis/facts/model/sbf/periodic.v	prm_sbf_valid_aux_2
3	prosa/rt-proofs-v0.6/implementation/refinements/fast_search_space_computation.v	task_search_space_subset
3	prosa/results/rta/arm/fp/limited_preemptive.v	uniprocessor_response_time_bound_limited_fp
3	prosa/util/nondecreasing.v	distances_iota_filtered
3	prosa/classic/analysis/apa/bertogna_edf_theory.v	bertogna_edf_interference_by_different_tasks
3	prosa/classic/analysis/global/jitter/workload_bound_n_k_covers_middle_jobs.v	workload_bound_n_k_covers_middle_jobs
3	prosa/classic/implementation/global/basic/schedule.v	scheduler_completed_jobs_dont_execute
3	prosa/classic/model/schedule/uni/limited/fix_priority/response_time_bound.v	instantiated_task_interference_is_bounded
3	prosa/util/nondecreasing.v	distance_between_neighboring_elements_le_max_distance_in_seq
4	prosa/analysis/abstract/restricted_supply/bounded_bi/edf.v	longest_bi_with_pi_bound_is_valid
4	prosa/results/rta/ovh/edf/limited_preemptive.v	uniprocessor_response_time_bound_limited_edf
4	prosa/util/nondecreasing.v	max_distance_in_nontrivial_seq_is_positive
4	prosa/analysis/facts/busy_interval/existence.v	busy_interval_is_bounded
4	prosa/results/rta/ovh/fp/limited_preemptive.v	uniprocessor_response_time_bound_limited_fp
4	prosa/classic/analysis/apa/bertogna_edf_theory.v	bertogna_edf_all_cpus_in_affinity_busy
4	prosa/classic/analysis/global/jitter/bertogna_fp_theory.v	bertogna_fp_interference_by_different_tasks
4	prosa/classic/analysis/uni/susp/sustainability/singlecost/reduction_properties.v	sched_susp_highercost_depends_only_on_prefix
4	prosa/classic/model/schedule/uni/limited/edf/nonpr_reg/response_time_bound.v	priority_inversion_is_bounded
4	prosa/util/nondecreasing.v	max_distance_in_nontrivial_seq_is_positive

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TABLE I (continued)

Level	File	Theorem Name
5	prosa/analysis/abstract/restricted_supply/iw_readiness.v	pending_hep_job_exists_inside_busy_interval
5	prosa/analysis/facts/busy_interval/existence.v	pending_hp_job_exists
5	prosa/implementation/refinements/EDF/fast_search_space.v	search_space_subset_EDF
5	prosa/classic/analysis/apa/bertogna_fp_comp.v	fp_analysis_yields_response_time_bounds
5	prosa/classic/analysis/global/basic/bertogna_fp_comp.v	fp_analysis_yields_response_time_bounds
5	prosa/classic/analysis/uni/susp/sustainability/allcosts/reduction_properties.v	sched_new_work_conserving
5	prosa/classic/implementation/apa/schedule.v	scheduler_priority
5	prosa/classic/model/schedule/global/jitter/platform.v	work_conserving_eq_work_conserving_count
5	prosa/classic/model/schedule/uni/limited/fix_priority/nonpr_reg/concrete_models/response_time_bound.v	uniprocessor_response_time_bound_fp_with_floating_nonpreemptive_regions
5	prosa/classic/analysis/global/parallel/bertogna_fp_comp.v	fp_analysis_yields_response_time_bounds
6	prosa/classic/analysis/apa/bertogna_edf_comp.v	bertogna_edf_comp_rt_grows_too_much
6	prosa/classic/analysis/global/basic/bertogna_edf_comp.v	bertogna_edf_comp_rt_grows_too_much
6	prosa/classic/analysis/global/jitter/interference_bound_edf.v	interference_bound_edf_holds_for_single_job_that_completes_on_time
6	prosa/classic/analysis/global/parallel/bertogna_edf_theory.v	bertogna_edf_interference_on_all_cpus
6	prosa/classic/analysis/uni/susp/dynamic/jitter/jitter_schedule_service.v	jitter_reduction_inductive_step_case2
6	prosa/classic/model/schedule/uni/limited/abstract_RTA/abstract_rta.v	job_completed_by_arrival_plus_R_2
6	prosa/classic/model/schedule/uni/limited/jlfp_instantiation.v	instantiated_quiet_time_equivalent_edf_quiet_time
6	prosa/classic/model/schedule/uni/limited/platform/priority_inversion_is_bounded.v	not_quiet_implies_exists_scheduled_hp_job_at_preemption_point
6	prosa/classic/model/schedule/uni/susp/build_suspension_table.v	suspension_duration_matches_predicate_up_to_t_max
6	prosa/classic/model/schedule/uni/susp/suspension_intervals.v	cumulative_suspension_eq_total_suspension
7	prosa/classic/analysis/apa/bertogna_fp_theory.v	bertogna_fp_interference_in_non_full_processors
7	prosa/classic/analysis/global/parallel/bertogna_edf_comp.v	bertogna_edf_comp_iteration_preserves_order
7	prosa/classic/analysis/uni/basic/tdma_wert_analysis.v	formula_sched_St
7	prosa/classic/model/schedule/uni/limited/abstract_RTA/abstract_seq_rta.v	bound_for_cumulative_job_interference_actual
7	prosa/classic/model/schedule/uni/limited/platform/priority_inversion_is_bounded.v	preemption_time_exists

TABLE II: Test data in Prosa

Level	File	Theorem Name
1	prosa/analysis/abstract/abstract_rta.v	uniprocessor_response_time_bound
1	prosa/analysis/facts/behavior/arrivals.v	any_ready_job_is_pending
1	prosa/model/task/arrival/sporadic_as_curve.v	sporadic_arrival_curve_valid
1	prosa/implementation/refinements/EDF/fast_search_space.v	EDF_ss_generalize_FP_ss
1	prosa/results/transfer_schedulability/criterion.v	ref_transfer_schedulability_criterion_sufficiency
1	prosa/classic/analysis/global/parallel/bertogna_edf_theory.v	bertogna_edf_specific_bound_holds
1	prosa/classic/analysis/uni/basic/fp_rta_comp.v	taskset_schedulable_by_fp_rta
1	prosa/classic/model/arrival/basic/arrival_bounds.v	sporadic_task_arrival_bound
1	prosa/classic/implementation/apa/schedule.v	scheduler_mapping_respects_affinity
1	prosa/classic/util/minmax.v	seq_max_nat_computes_max
2	prosa/analysis/abstract/busy_interval.v	busy_interval_has_uninterrupted_service
2	prosa/analysis/abstract/IBF/task.v	task_IBF_implies_job_IBF
2	prosa/analysis/abstract/restricted_supply/search_space/fifo_fixpoint.v	soln_abstract_response_time_recurrence
2	prosa/implementation/refinements/fast_search_space_computation.v	multiple_of_horizon_in_approx_ss
2	prosa/results/transfer_schedulability/criterion.v	slackless_interval_existence
2	prosa/classic/analysis/apa/bertogna_edf_theory.v	bertogna_cirinei_response_time_bound_edf
2	prosa/classic/analysis/global/jitter/bertogna_edf_theory.v	bertogna_edf_workload_bounds_interference
2	prosa/classic/analysis/uni/basic/tdma_wcrt_analysis.v	exists_WCRT
2	prosa/classic/analysis/uni/susp/dynamic/oblivious/fp_rta.v	suspension_oblivious_fp_rta_implies_schedulability
2	prosa/classic/model/schedule/global/transformation/construction.v	service_dependent_schedule_construction
3	prosa/analysis/abstract/busy_interval.v	busy_interval_is_bounded
3	prosa/analysis/facts/behavior/arrivals.v	arrivals_between_sorted
3	prosa/implementation/refinements/EDF/nonpreemptive_sched.v	sched_nonpreemptive_next
3	prosa/results/rta/ideal/elf/bounded_pi.v	instantiated_busy_intervals_are_bounded
3	prosa/util/search_arg.v	search_arg_extremum
3	prosa/classic/analysis/apa/bertogna_fp_theory.v	bertogna_fp_workload_bounds_interference
3	prosa/classic/analysis/global/parallel/interference_bound_edf.v	interference_bound_edf_n_k_covers_all_jobs
3	prosa/classic/analysis/uni/basic/fp_rta_theory.v	uniprocessor_response_time_bound_fp
3	prosa/classic/implementation/apa/schedule.v	scheduler_mapping_is_work_conserving
3	prosa/classic/model/schedule/uni/limited/abstract_RTA/reduction_of_search_space.v	representative_exists
4	prosa/analysis/abstract/ideal/iw_instantiation.v	quiet_time_ab_implies_quiet_time_cl
4	prosa/analysis/facts/behavior/service.v	incremental_service_during
4	prosa/analysis/facts/model/overheads/blackout_bound.v	fin_sched_changes_start_busy_pref_bounded_overheads_blackout
4	prosa/results/rta/ovh/edf/fully_nonpreemptive.v	uniprocessor_response_time_bound_fully_nonpreemptive_edf
4	prosa/util/unit_growth.v	slowed_subtraction_value_preservation
4	prosa/classic/analysis/apa/bertogna_edf_comp.v	bertogna_edf_comp_f_increases
4	prosa/classic/analysis/apa/bertogna_fp_theory.v	bertogna_fp_all_cpus_in_subaffinity_busy
4	prosa/classic/analysis/uni/basic/tdma_wcrt_analysis.v	formula_not_sched_St
4	prosa/classic/model/schedule/global/basic/constrained_deadlines.v	platform_cpus_busy_with_interfering_tasks
4	prosa/classic/util/div_mod.v	divSn_cases

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TABLE II (continued)

Level	File	Theorem Name
5	prosa/classic/analysis/apa/bertogna_edf_comp.v	edf_claimed_bounds_finds_fixed_point_of_list
5	prosa/classic/analysis/global/basic/bertogna_edf_comp.v	edf_claimed_bounds_finds_fixed_point_of_list
5	prosa/classic/analysis/global/basic/interference_bound_edf.v	interference_bound_edf_holds_for_single_job_that_completes_on_time
5	prosa/classic/analysis/global/jitter/bertogna_edf_comp.v	edf_claimed_bounds_finds_fixed_point_of_list
5	prosa/classic/analysis/global/jitter/bertogna_fp_comp.v	fp_analysis_yields_response_time_bounds
5	prosa/classic/analysis/global/parallel/bertogna_edf_comp.v	edf_claimed_bounds_finds_fixed_point_of_list
5	prosa/classic/analysis/uni/basic/tdma_wcrt_analysis.v	response_time_le_WCRT
5	prosa/classic/analysis/uni/susp/dynamic/jitter/jitter_schedule_properties.v	sched_jitter_respects_policy
5	prosa/classic/model/schedule/uni/limited/abstract_RTA/abstract_seq_rta.v	task_rbf_excl_tsk_bounds_task_workload_excl_j
5	prosa/classic/model/schedule/uni/limited/edf/nonpr_reg/concrete_models/response_time_bound.v	uniprocessor_response_time_bound_edf_with_floating_nonpreemptive_regions
6	prosa/classic/analysis/apa/bertogna_edf_theory.v	bertogna_edf_alpha'is_full
6	prosa/classic/analysis/global/basic/bertogna_edf_theory.v	bertogna_edf_interference_in_non_full_processors
6	prosa/classic/analysis/global/jitter/bertogna_edf_comp.v	bertogna_edf_comp_rt_grows_too_much
6	prosa/classic/analysis/global/parallel/bertogna_edf_comp.v	bertogna_edf_comp_rt_grows_too_much
6	prosa/classic/analysis/global/parallel/bertogna_fp_theory.v	bertogna_fp_all_cpus_are_busy
6	prosa/classic/model/schedule/uni/limited/busy_interval.v	busy_interval_is_bounded
6	prosa/classic/model/schedule/uni/limited/busy_interval.v	pending_hp_job_exists
6	prosa/classic/model/schedule/uni/limited/edf/response_time_bound.v	A_is_in_concrete_search_space
6	prosa/classic/model/schedule/uni/limited/jlfp_instantiation.v	instantiated_cumulative_interference_of_hep_jobs_equal_total_interference_of_hep_jobs
6	prosa/classic/model/schedule/uni/service.v	incremental_service_during
7	prosa/classic/analysis/apa/bertogna_edf_comp.v	bertogna_edf_comp_iteration_preserves_order
7	prosa/classic/analysis/global/jitter/bertogna_edf_comp.v	bertogna_edf_comp_iteration_preserves_order
7	prosa/classic/analysis/global/basic/bertogna_edf_comp.v	bertogna_edf_comp_iteration_preserves_order
7	prosa/classic/model/schedule/uni/limited/fixed_priority/nonpr_reg/concrete_models/response_time_bound.v	uniprocessor_response_time_bound_fp_with_fixed_preemption_points
7	prosa/classic/model/schedule/uni/limited/edf/nonpr_reg/concrete_models/response_time_bound.v	uniprocessor_response_time_bound_edf_with_fixed_preemption_points